

CrabTransformer Training Report

Data: REAL | Scenario: REAL | Loss: Tweedie

— Real Data —

Source: Eastern Bering Sea Snow Crab Survey
Grid: 50x50 (25km cells, SPDE interpolation)
Valid ocean cells: 886
Bootstraps: 100 (subsampling from ~349 stations)

— Training Configuration —

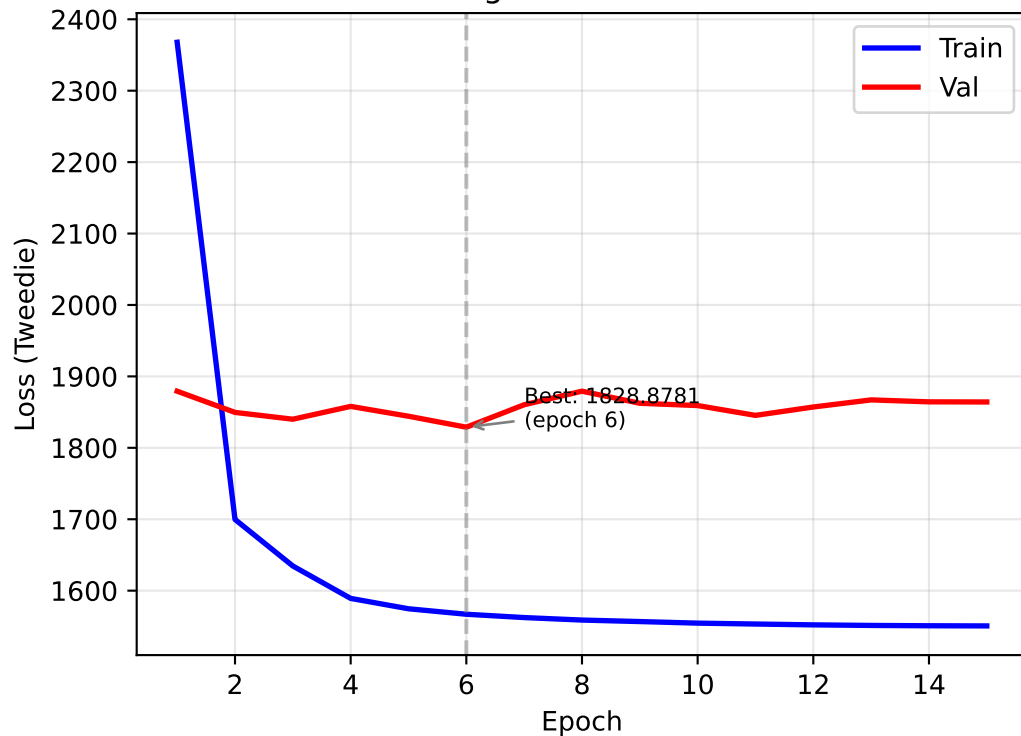
epochs_run: 15 best_val_epoch: 6
best_val_loss: 1828.8781 final_train_loss: 1550.6445
bias_correction (MSE): 1.0000

— Model Architecture —

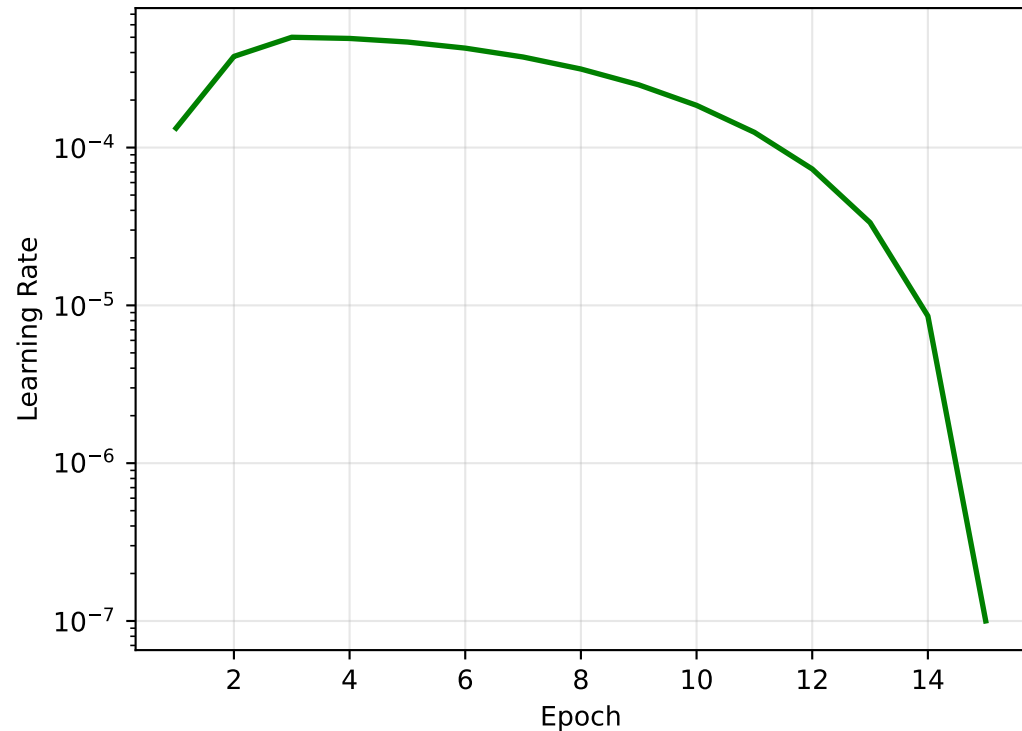
embed_dim: 128 num_heads: 4 num_layers: 3
d_ff: 256 dropout: 0.3 patch_size: 5
train/val/test years: 24/8/4

Training Curves: REAL | Tweedie

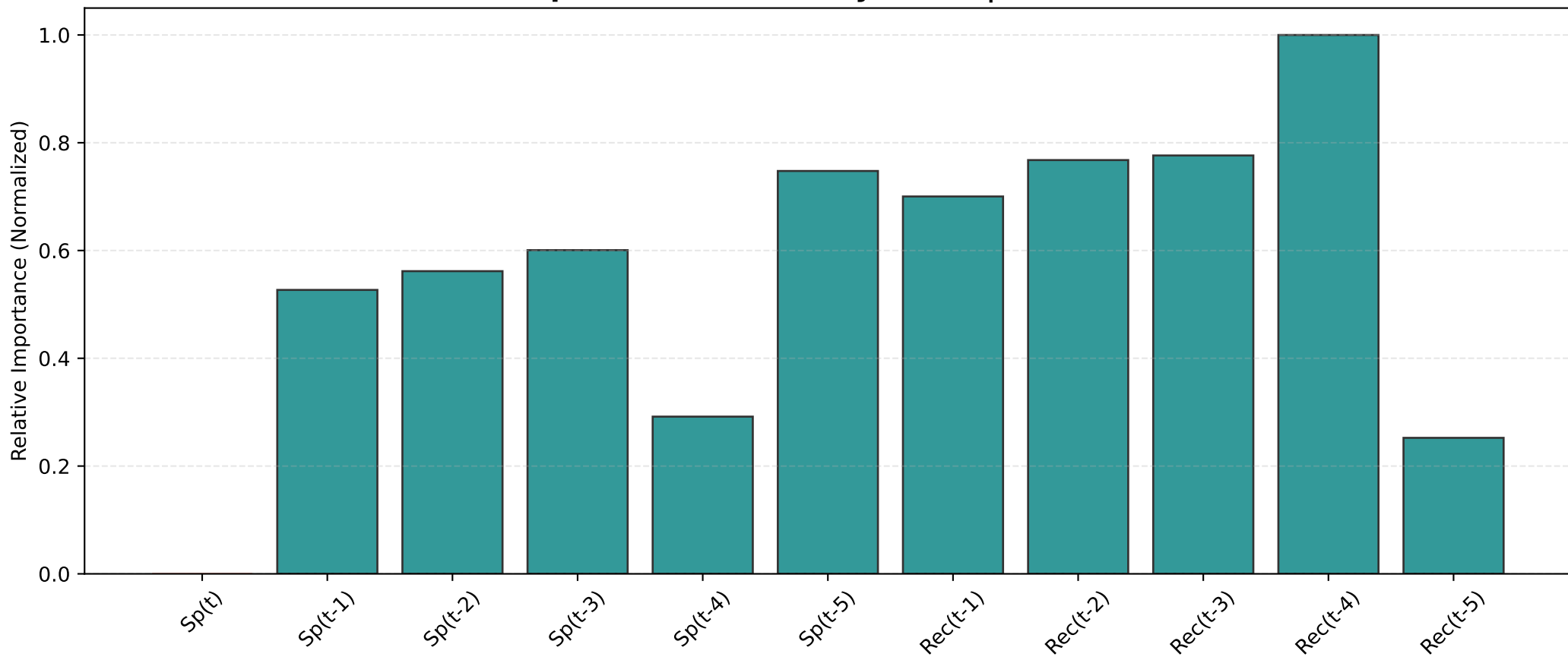
Training & Validation Loss



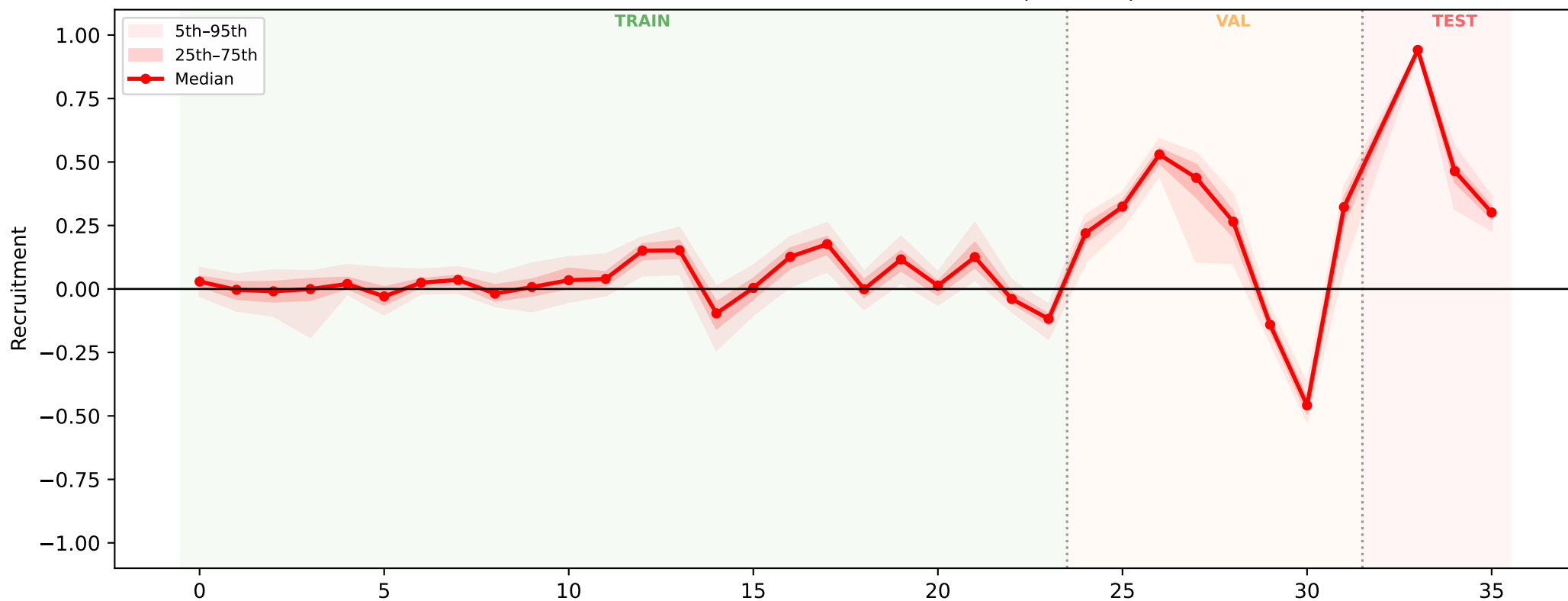
OneCycleLR Schedule



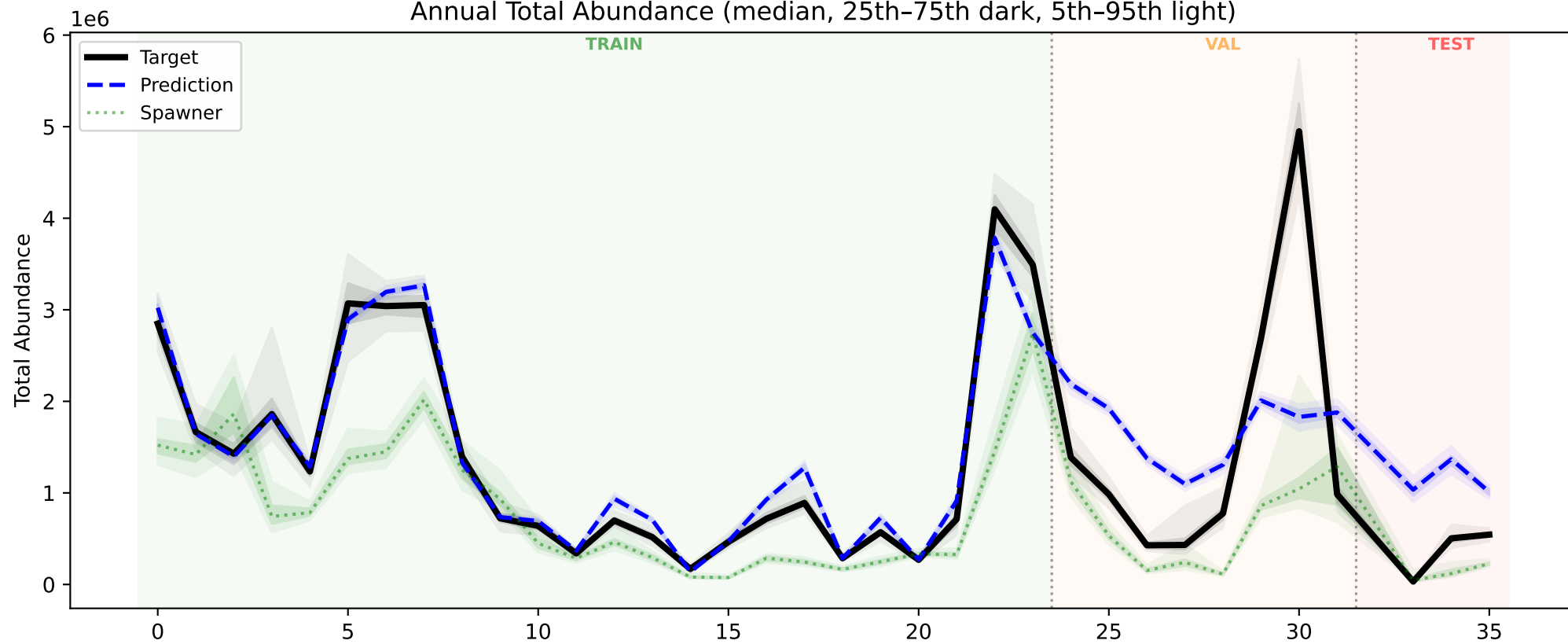
Input Channel Priority: REAL | Tweedie



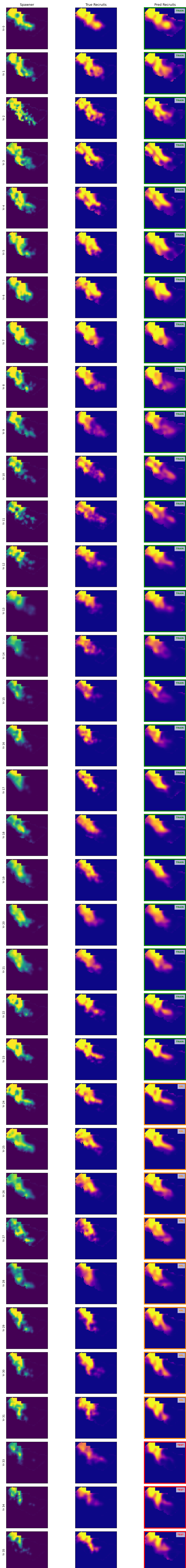
Annual Recruitment Deviations: REAL (Tweedie)



Annual Total Abundance (median, 25th-75th dark, 5th-95th light)



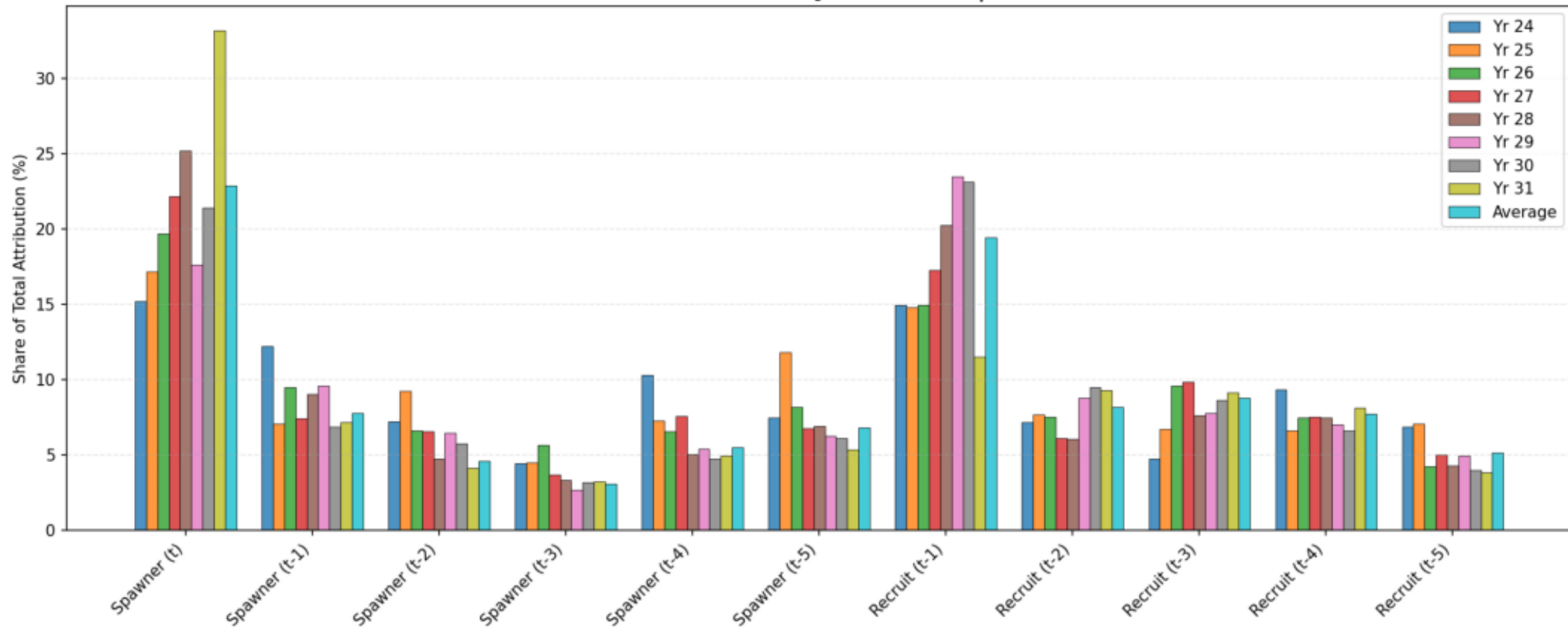
Spatial Trajectory: REAL | Tweedie



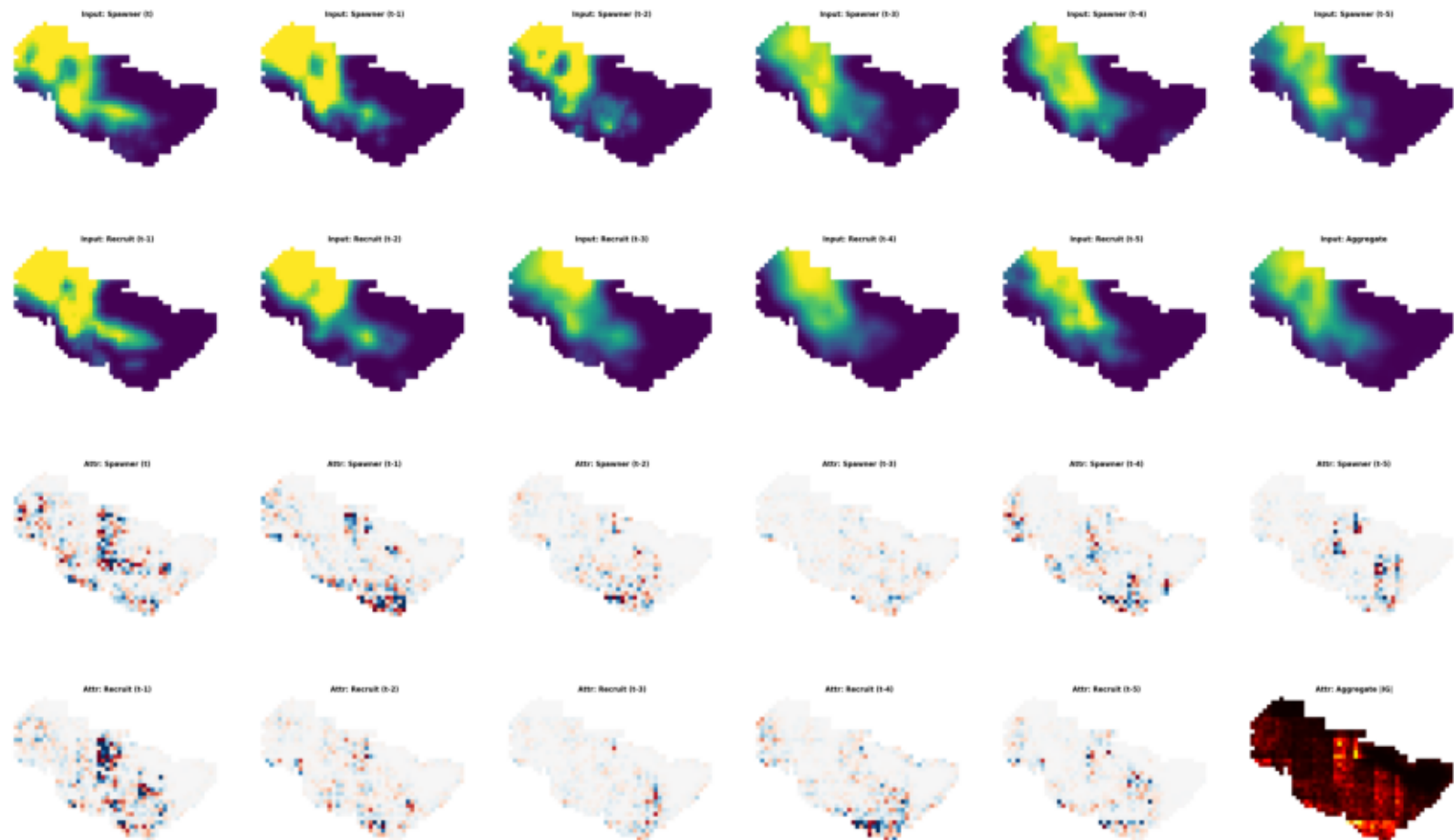
Year AVG — REAL | Tweedie
 Top: Mean Input (log-space) | Bottom: Attribution (red=+recruit, blue=-recruit)



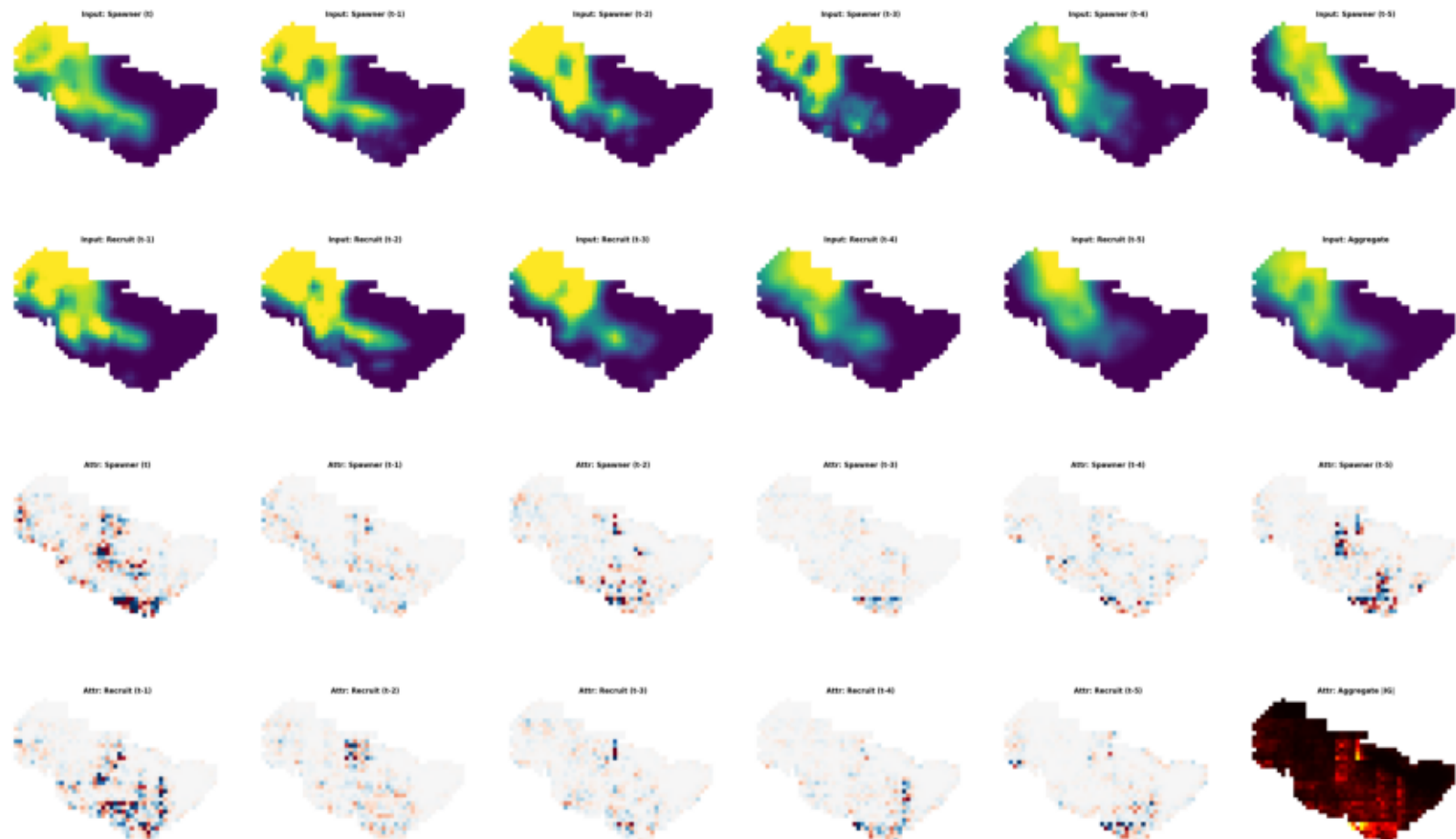
Channel Attribution by Year: REAL | Tweedie



Year 24 — REAL | Tweedie
 Top: Mean Input (log-space) | Bottom: Attribution (red=+recruit, blue=-recruit)



Year 25 — REAL | Tweedie
 Top: Mean Input (log-space) | Bottom: Attribution (red=+recruit, blue=-recruit)



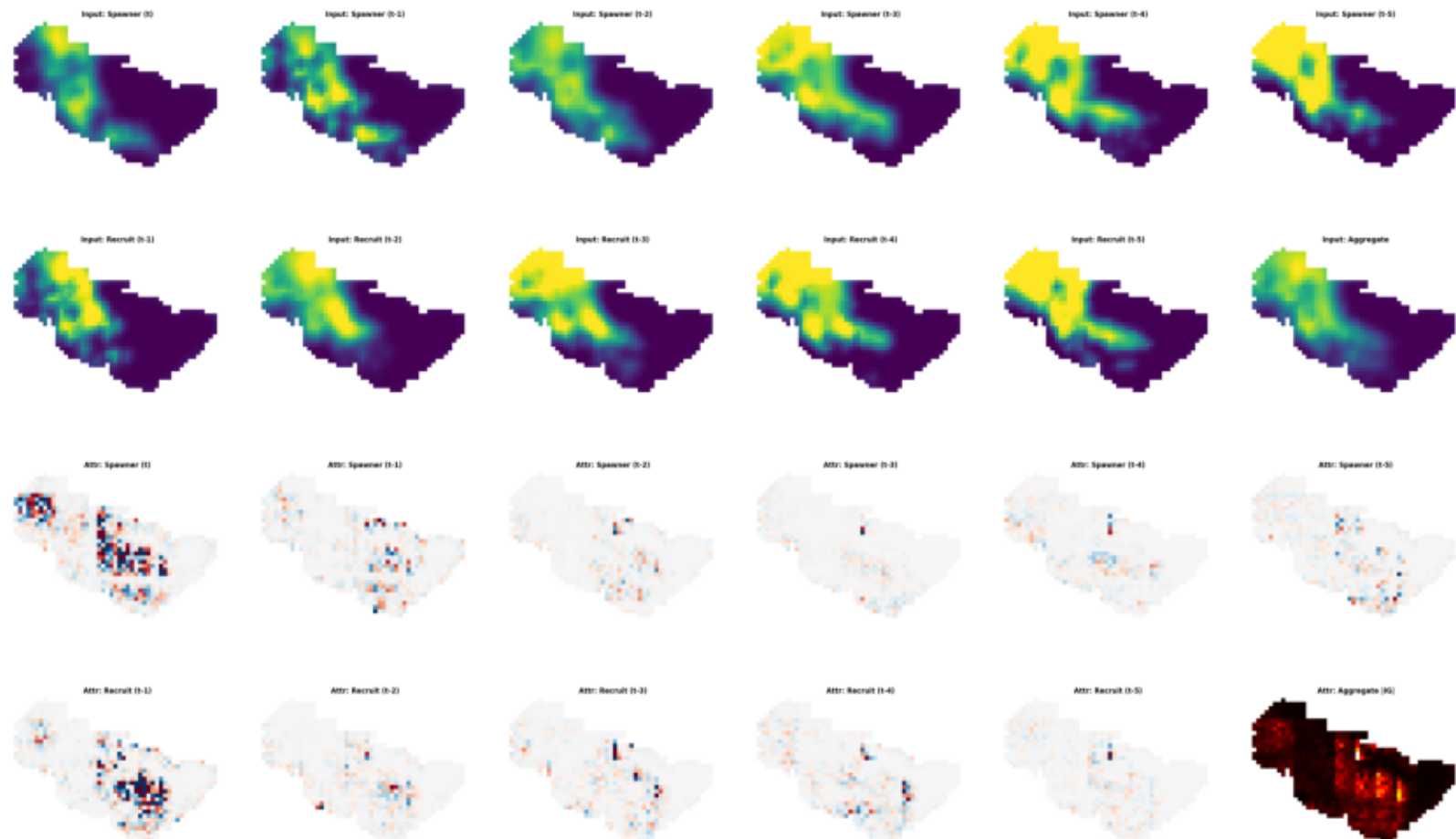
Year 26 — REAL | Tweedie
 Top: Mean Input (log-space) | Bottom: Attribution (red=+recruit, blue=-recruit)



Year 27 — REAL | Tweedie
 Top: Mean Input (log-space) | Bottom: Attribution (red=+recruit, blue=-recruit)



Year 28 — REAL | Tweedie
 Top: Mean Input (log-space) | Bottom: Attribution (red=+recruit, blue=-recruit)



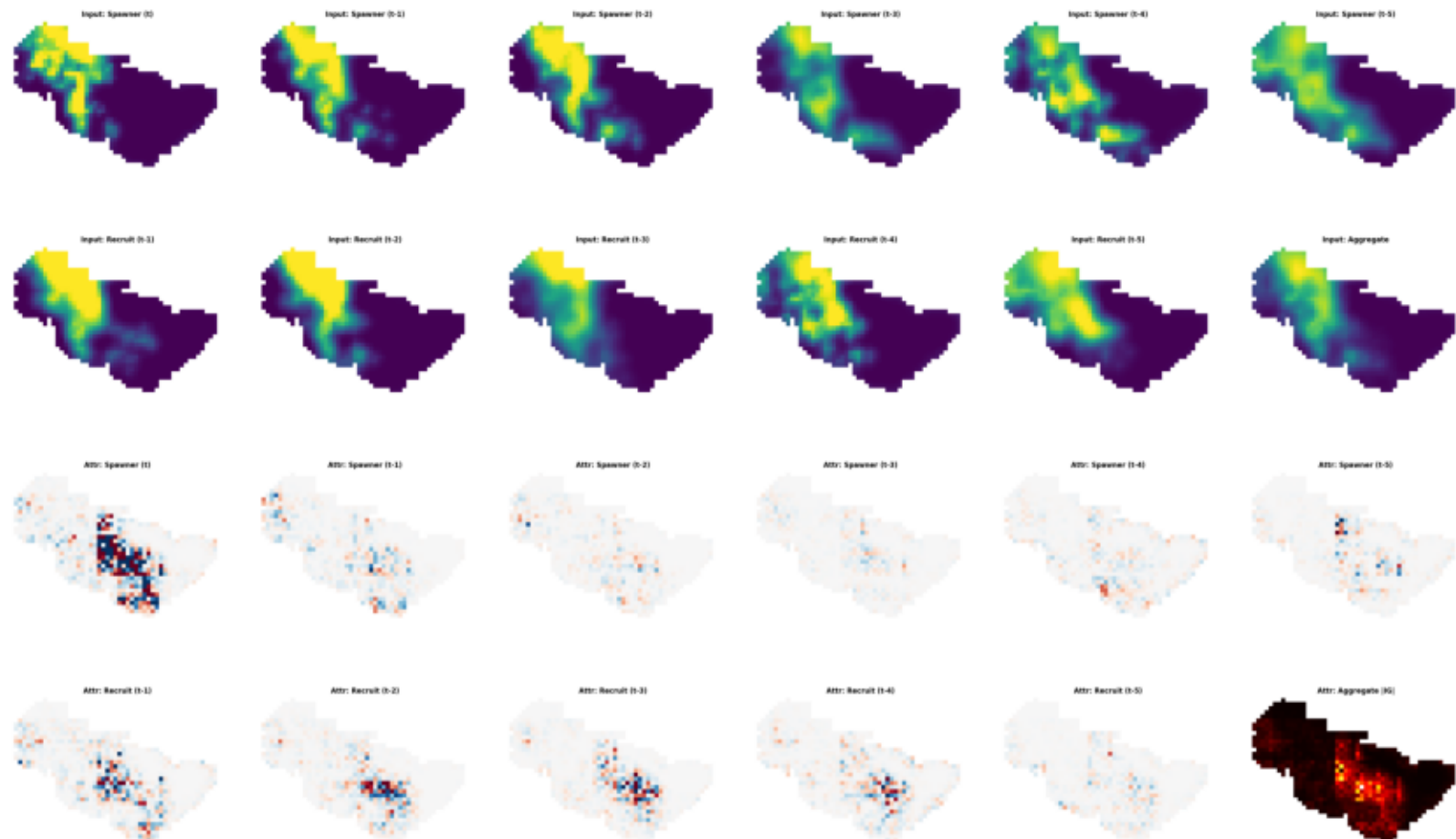
Year 29 — REAL | Tweedie
 Top: Mean Input (log-space) | Bottom: Attribution (red=+recruit, blue=-recruit)



Year 30 — REAL | Tweedie
 Top: Mean Input (log-space) | Bottom: Attribution (red=+recruit, blue=-recruit)



Year 31 — REAL | Tweedie
 Top: Mean Input (log-space) | Bottom: Attribution (red=+recruit, blue=-recruit)



Summary Statistics: REAL | Tweedie

Phase	Pred_Mean	Tgt_Mean	Pred_SD	Pred_P50	Pred_P95	Abundance	MAE	Spearman
TRAIN	1639.0	1612.8	5680.9	0.0	10402.3	1.016	479.5	0.942
VAL	1920.2	1801.5	5133.1	0.0	14598.6	1.066	1767.0	0.877
TEST	1278.2	406.3	4111.8	0.0	9444.6	3.146	944.2	0.860

Detailed Metrics (R-style): REAL | Tweedie

Phase	RMSE	RMSE_unbias	Bias	Bias_%	Spearman	Pearson	Pct_Error	Zero_F1	TopK_Acc	TopK_Jaccard	Quant_Err	Skill_MSE	Skill_MAE
TRAIN	1184.3	1169.3	26.2	2.2%	0.950	0.972	6.8%	0.950	0.911	0.838	0.33	0.908	0.834
VAL	5474.4	5335.5	118.7	5.5%	0.882	0.682	69.8%	0.913	0.682	0.531	0.80	-1.252	0.146
TEST	3122.2	2996.8	871.9	7.6%	0.871	0.711	1194.9%	0.934	0.746	0.603	0.64	-553.689	-6.738

Appendix A: Summary Metrics

APPENDIX A: Table Metric Definitions

Data type: REAL
Split: 24 train / 8 val / 4 test years

Spatial masking: All metrics computed ONLY over valid ocean cells (~886) in the 50x50 grid. Padding cells are excluded from all computations.

All metrics computed on ORIGINAL DENSITY SCALE (after expm1 and bias correction for MSE). Zero threshold = 14 (smallest nonzero observed value): predictions AND observations below 14 are set to 0 before ALL metric computation. Both Page 7 and Page 8 apply this threshold.

Summary Table (Page 7)

Computed per bootstrap sample, then averaged across all samples in phase.

Pred_Mean, Tgt_Mean: Arithmetic mean of valid cell values.
Pred_SD: Standard deviation of valid cell values.
Pred_P50, Pred_P95: Median and 95th percentile.
Abundance: $\text{sum}(\text{pred}) / \text{sum}(\text{target})$. 1.0 = perfect.
MAE: $\text{mean}(|\text{pred} - \text{target}|)$ per valid cell, averaged across samples.
Spearman: Rank correlation over valid cells, averaged across samples.

Appendix A (cont): Detailed Metrics

APPENDIX A (cont): Detailed Metrics (Page 8)

Per-year: average spatial field across bootstraps, then compute metrics on that mean field over valid cells. Average across years per phase.

RMSE: $\sqrt{\text{mean}((\text{obs} - \text{pred})^2)}$ over valid cells.

RMSE_unbiased: RMSE after subtracting mean bias.

Bias: $\text{mean}(\text{pred} - \text{obs})$. Positive = overprediction.

Bias_%: $\text{bias}^2 / \text{rmse}^2 * 100$.

Spearman / Pearson: Rank and linear correlation.

Pct_Error: $(\text{sum}(\text{pred}) - \text{sum}(\text{obs})) / \text{sum}(\text{obs}) * 100$.

Zero_F1: F1 for zero-classification (threshold = 14).

TopK_Accuracy / Jaccard: Overlap of top-10% density cells.

Quantile_Mean_Error: Mean |bin diff| across 10 quantile bins.

Skill_MSE / Skill_MAE: $1 - \text{error}/\text{climatology}$. Positive = beats mean.

Appendix B: Plot Methodology

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Page 2: Training Curves

MSE loss in log-space; Tweedie on original scale.

OneCycleLR: max_lr/10 -> max_lr (10%), cosine to max_lr/1000.

Page 3: Channel Importance

mean(|weights|) of each channel's projection. Static, not causal.

Page 4: RecDev & Abundance

RecDev = (pred - target) / (target + pred). Sums over valid cells.

Ribbons: 5th-95th and 25th-75th percentiles across bootstraps.

Page 5: Spatial Trajectory

Log-space grids, first bootstrap, shared colorscale 0-8.

Page 6: Integrated Gradients

Pre-computed, averaged across bootstraps per year.

Output function sums over valid cells only (real data).

Bias Correction (MSE only)

pred = expm1(pred_log) * exp(sigma^2 / 2)

Tweedie: no correction needed.